



جامعة نيويورك أبوظبي
NYU ABU DHABI

Advanced Circuits

ENGR-UH 2311 | Spring I - 2025

Experiment: **Lab 5**

Date: **25 / 02 / 2025**

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Lab 5: Transient Simulation of circuits involving RLC in Cadence and the Power conservation in the circuits

Objectives:

The main objective of this lab is to learn the Cadence environment and create schematics in Cadence, and define parameters of the electrical components. The lab includes setting up the simulation environment and defining its parameters, running simulations and analyzing circuit behaviors, running the transient response of the circuits, choosing the variables from the schematic, plotting different variables and properties of the plots, and exporting high-resolution images of the schematics and plots.

Diagram:

Circuit 1:

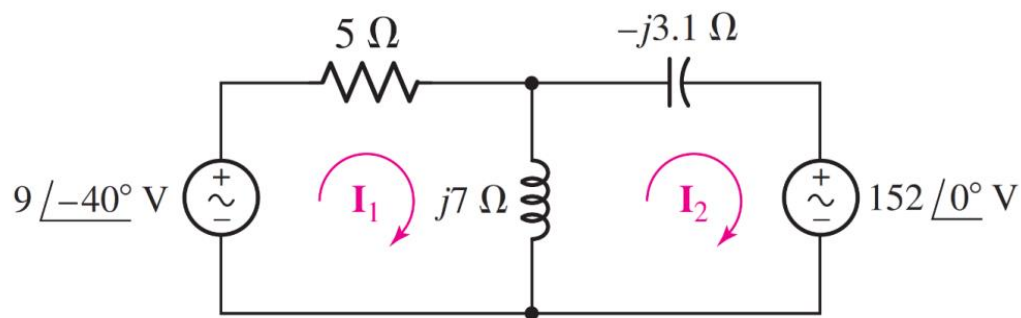


Figure 1: Circuit 1 RLC Schematic with Voltage Sources of different Phases

Circuit 2: Transient Analysis

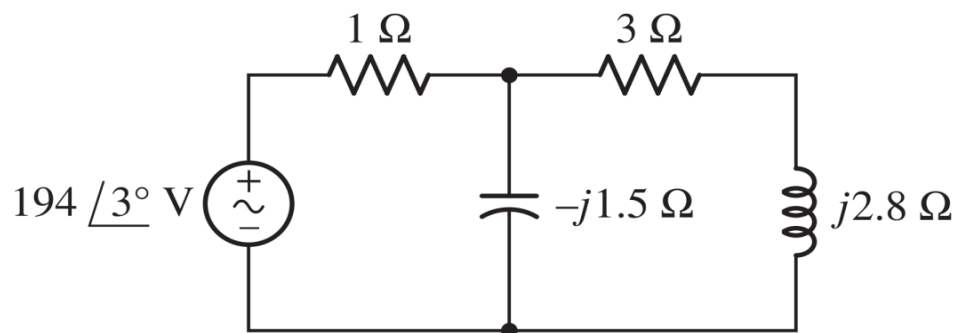


Figure 2: Circuit 2 RLC Schematic and a Voltage Source

Procedure:

Circuit 1: RLC Circuit with Sources with Phase Differences

1. An RLC circuit was built in Cadence, consisting of a resistor ($R = 5\Omega$), an inductor ($Z = j7\Omega$), a capacitor ($Z = -j3.1\Omega$) and 2 voltage sources as shown in the schematic in Figure 1 and the frequency of 1kHz was assumed.
2. Voltage sources with specific phase angles were configured appropriately to create phase differences.
 - Voltage source 1: $9\angle-40^\circ$ ($9 \cos(2000\pi t - 40^\circ)$) V was converted to its sine equivalent of $9 \sin(2000\pi t + 50^\circ)$ V and was configured appropriately to create phase differences as required.

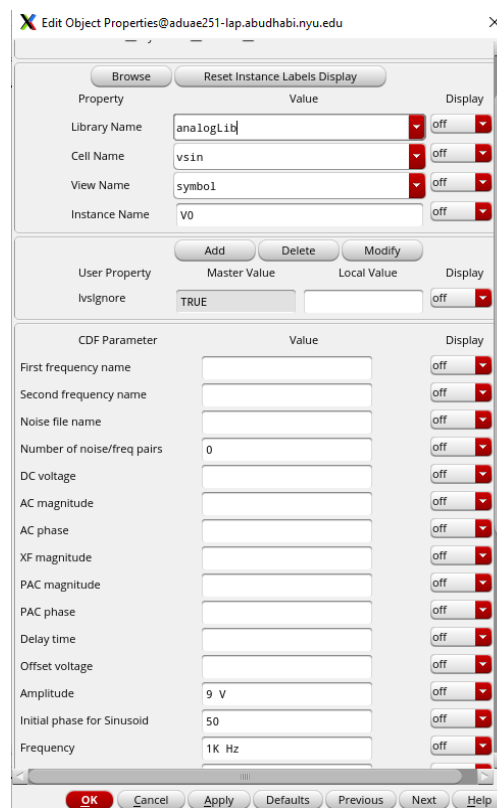


Figure 3: Circuit 1 Voltage Source 1 (Left)

- Voltage source 2: $152\angle 0^\circ$ ($(152 \cos(2000\pi t))$) V was converted to its sine equivalent of $(9 \sin(2000\pi t + 90^\circ))$ V and was configured appropriately to create phase differences as required.

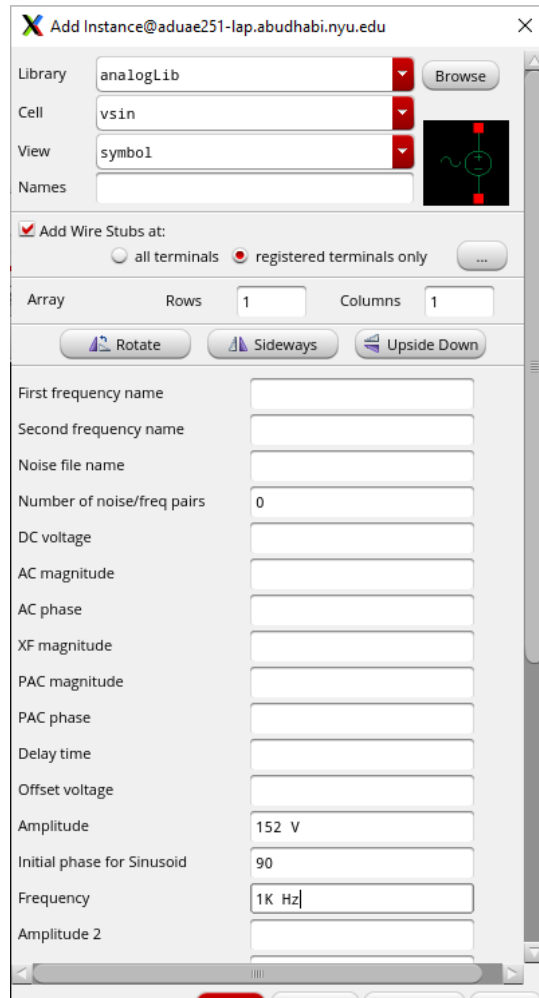


Figure 4: Circuit 1 Voltage Source 2 (Right)

3. The Cadence Calculator was used to calculate the Power supplied/absorbed by the elements as shown below.

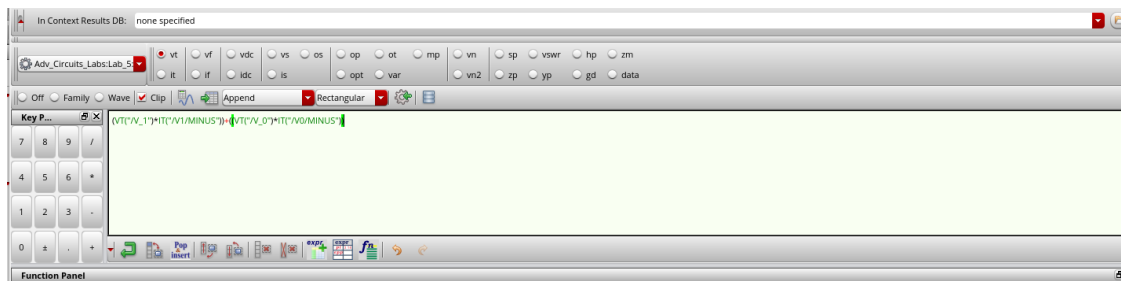


Figure 5: Total Power Supplied by the Voltage Sources in Cadence Calculator

- The transient analysis was performed and the average power absorbed by each passive element, and the average power supplied by each source was determined.

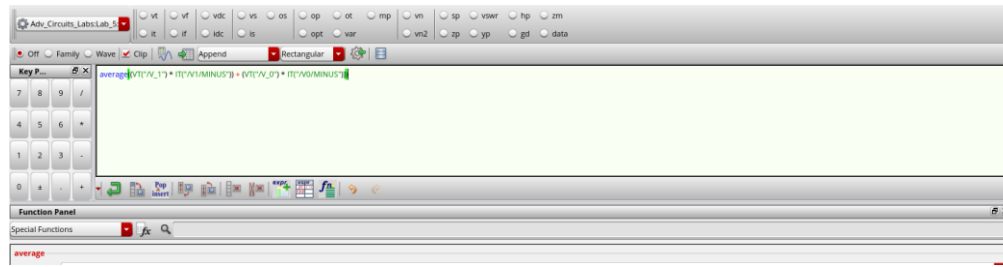


Figure 6: Total Average Power Supplied Calculation using Cadence Calculator

- The instantaneous power absorbed and supplied by all the elements of the circuit was plotted.
- The simulation results were examined to verify theoretical expectation that:

$$\text{Total supplied average power} = \text{Total absorbed average power}$$

Circuit 2: RLC Circuit with a Voltage Source

- An RLC circuit was implemented in Cadence, consisting of 2 resistors ($R = 1\Omega$ and $R = 3\Omega$ each), an inductor ($Z = j2.8\Omega$), a capacitor ($Z = -j1.5\Omega$), and a voltage source.
- A voltage source with a specific phase angle was configured appropriately to create phase differences.
 - Voltage source: $194\angle 3^\circ$ ($194 \cos(2000\pi t + 3^\circ)$) V was converted to its sine equivalent of $194 \sin(2000\pi t + 93^\circ)$ V and was configured appropriately to create phase differences as required.

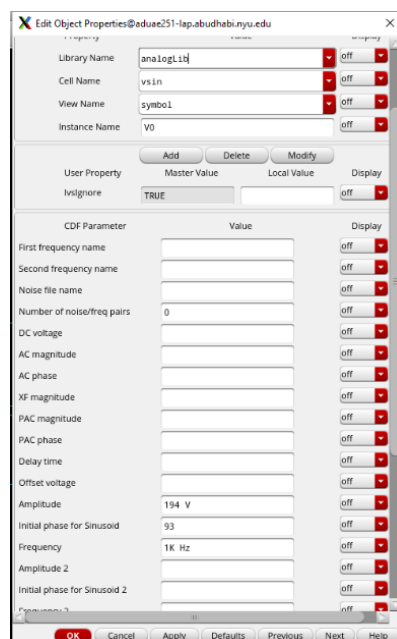


Figure 7: Circuit 2 Voltage Source

3. The transient analysis was performed and the average power absorbed by each passive element, and the average power supplied by each source was determined.

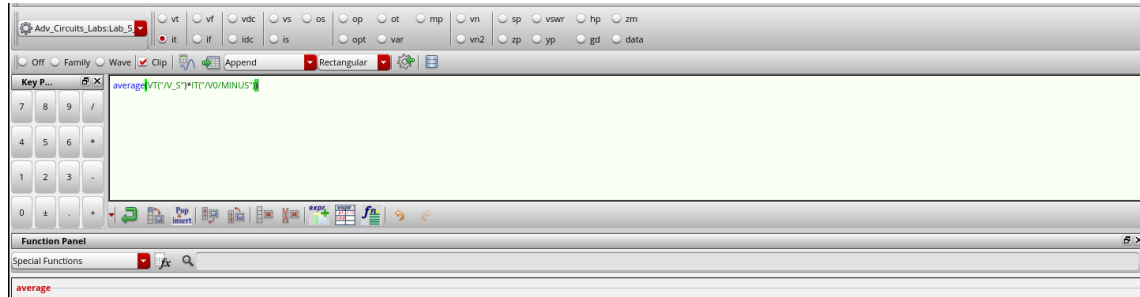


Figure 8: Total Average Power Supplied Calculation using Cadence Calculator

4. The instantaneous power absorbed and supplied by all the elements of the circuit was plotted.
5. The simulation results were examined to verify theoretical expectation that:

$$\text{Total supplied average power} = \text{Total absorbed average power}$$

Results & Observations

Circuit 1:

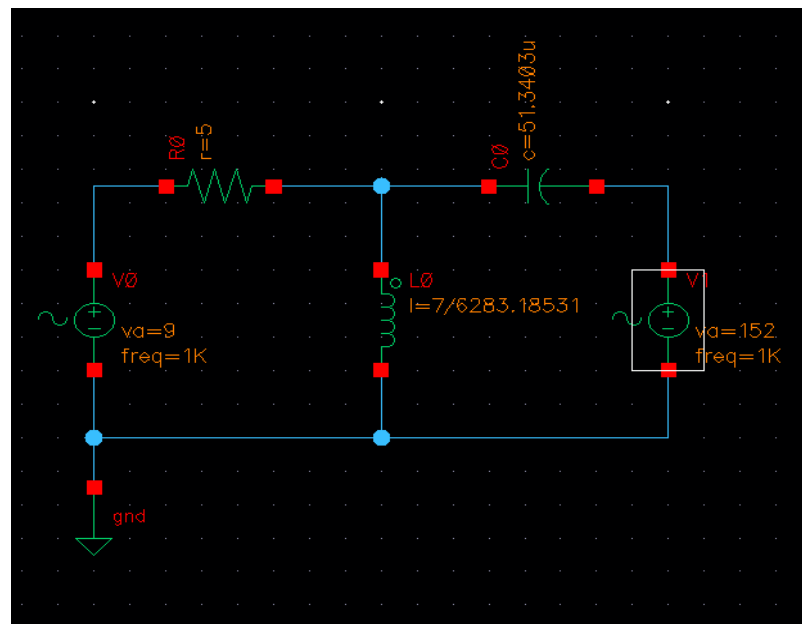


Figure 9: Circuit Schematic 1 Cadence Implementation

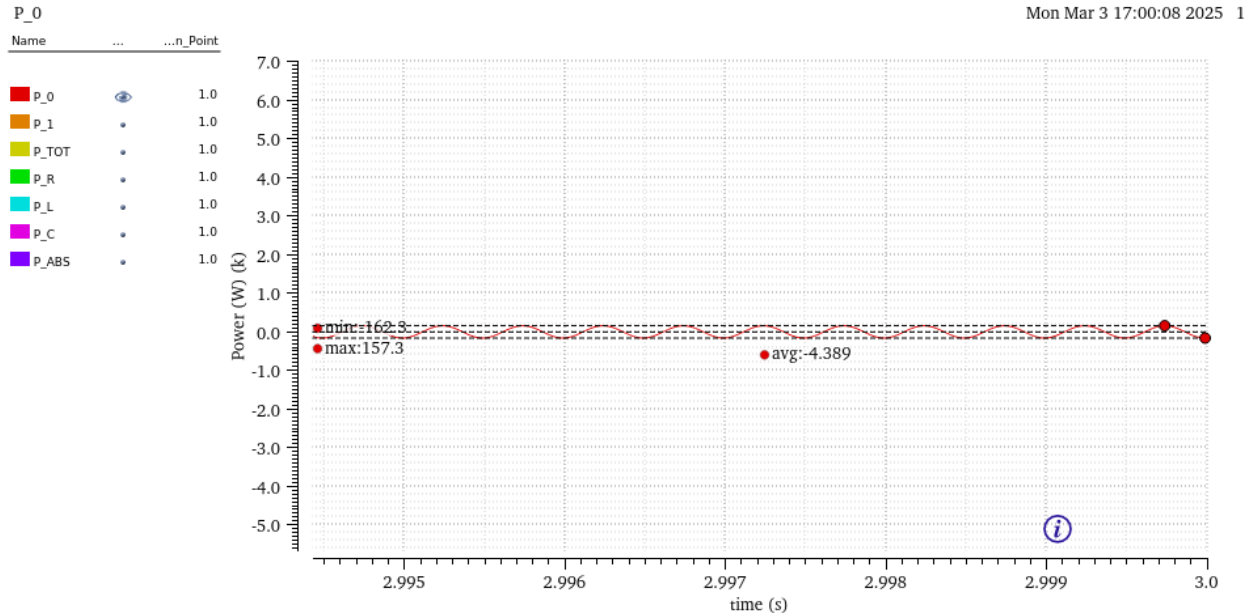


Figure 10: Plot of Power for Voltage Source 1 (Left)

The magnitude of the average power supplied by Voltage source 1 ($9 \angle -40^\circ$ V) was found to be 4.389 W as shown in Figure 10 above.

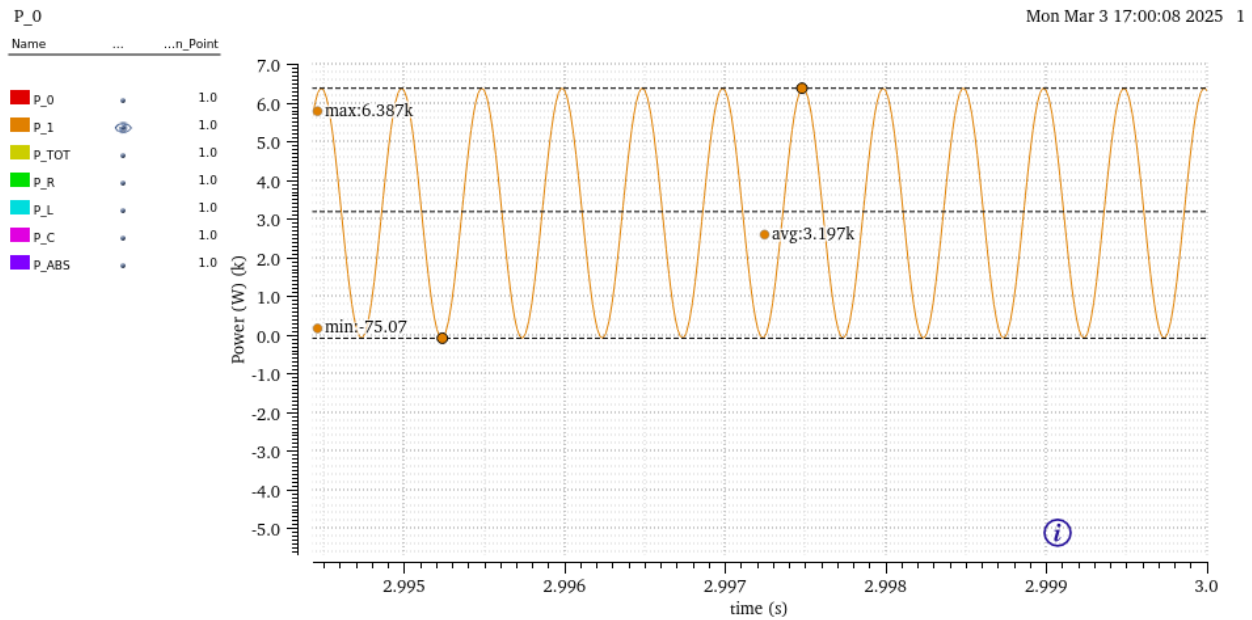


Figure 11: Plot of Power for Voltage Source 2 (Right)

The magnitude of the average power supplied by Voltage source 1 ($152 \angle 0^\circ$ V) was found to be 3.197 kW as shown in Figure 11 above.

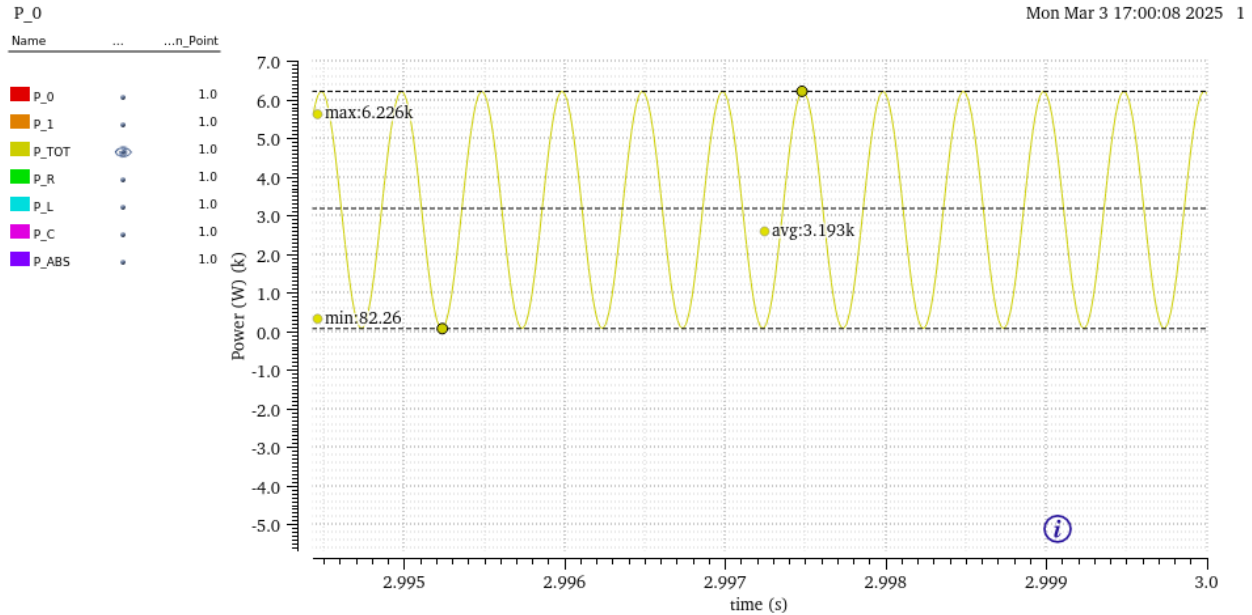


Figure 12: Plot of Total Power Supplied for the Two Voltage Sources

The magnitude of the average total power supplied by Voltage sources was found to be 3.193 kW as shown in Figure 12 above.

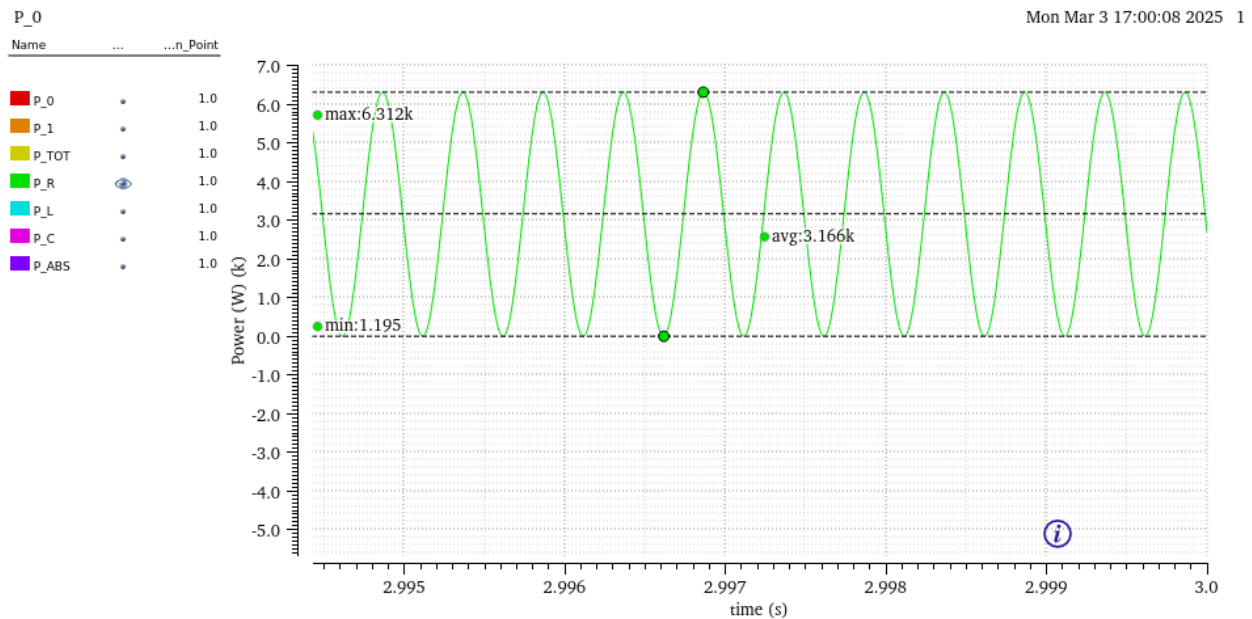


Figure 13: Plot of Power Absorbed by the Resistor

The magnitude of the average power absorbed by the resistor was found to be 3.166 kW as shown in Figure 13 above.

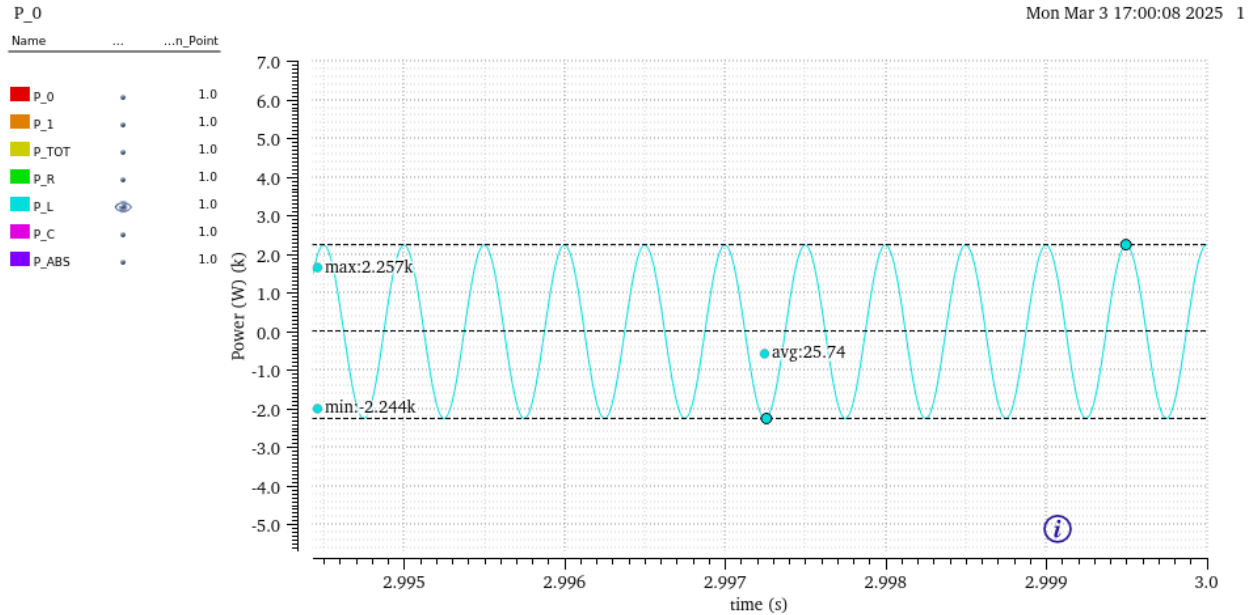


Figure 14: Plot of Power Absorbed by the Inductor

The magnitude of the average power absorbed by the inductor was found to be 25.74 W as shown in Figure 14 above.

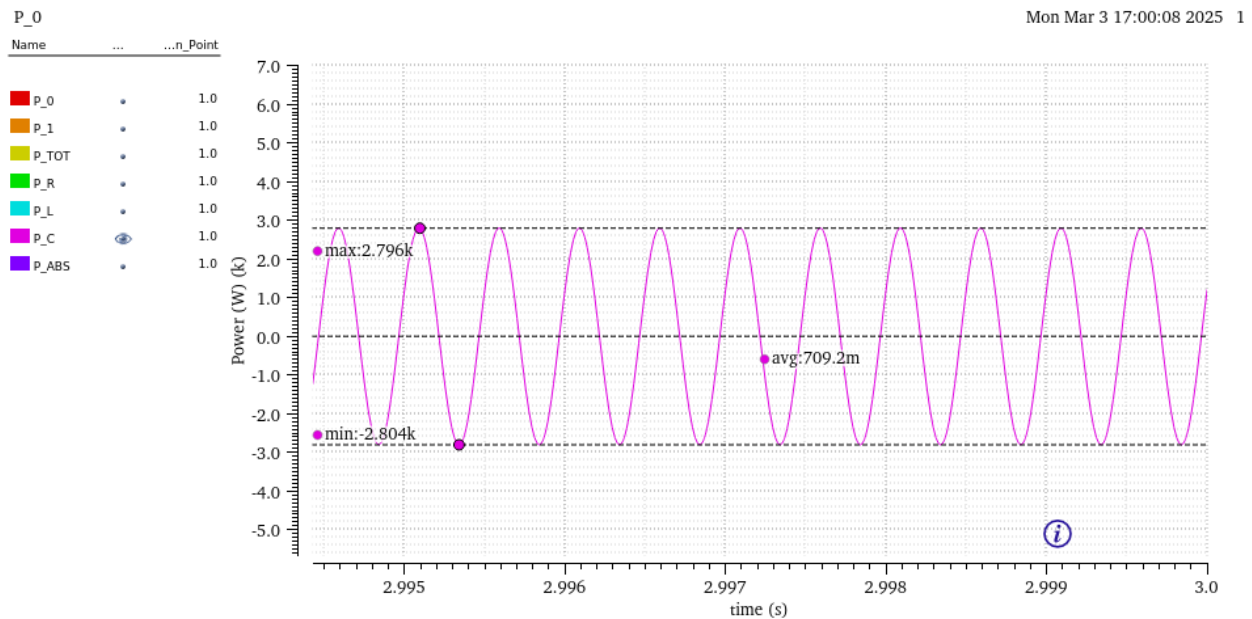


Figure 15: Plot of Power Absorbed by the Capacitor

The magnitude of the average power absorbed by the capacitor was found to be 709.2 mW as shown in Figure 15 above.

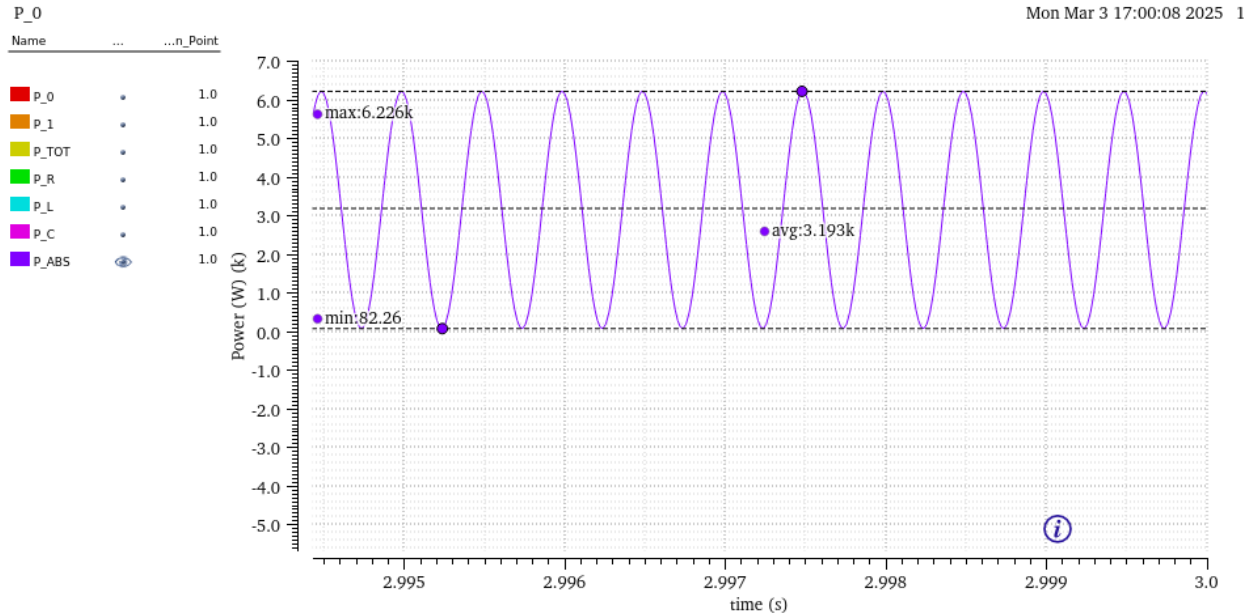


Figure 16: Plot of Total Power Absorbed by the Passive Elements

The magnitude of the total average power absorbed by the passive elements was found to be 3.193 kW as shown in Figure 16 above.

Comparing Figure 12 showing the *Total Average Power Supplied* and Figure 16 showing the *Total Average Power Absorbed*, it can be confirmed that the total average power absorbed and supplied are equal: 3.193 kW.

Circuit 2

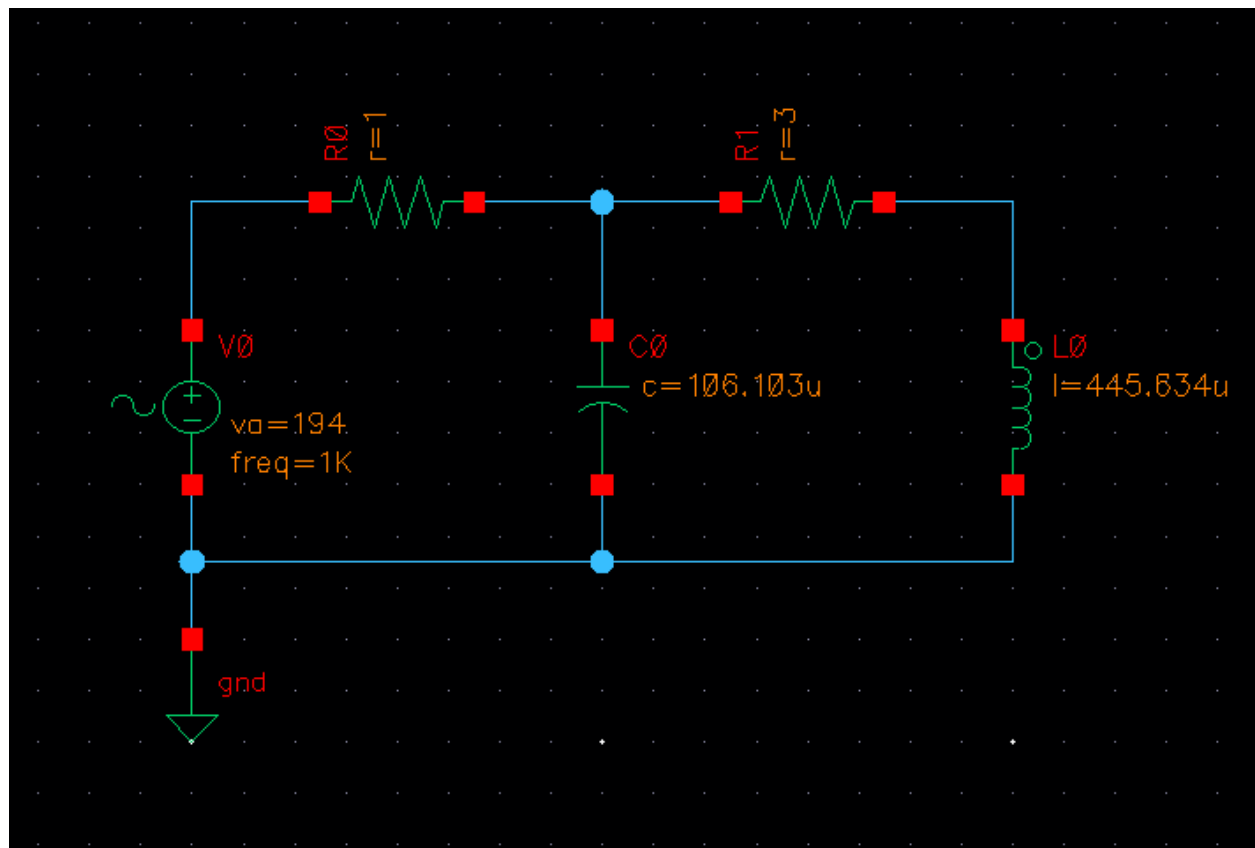


Figure 17: Circuit Schematic 2 Cadence Implementation

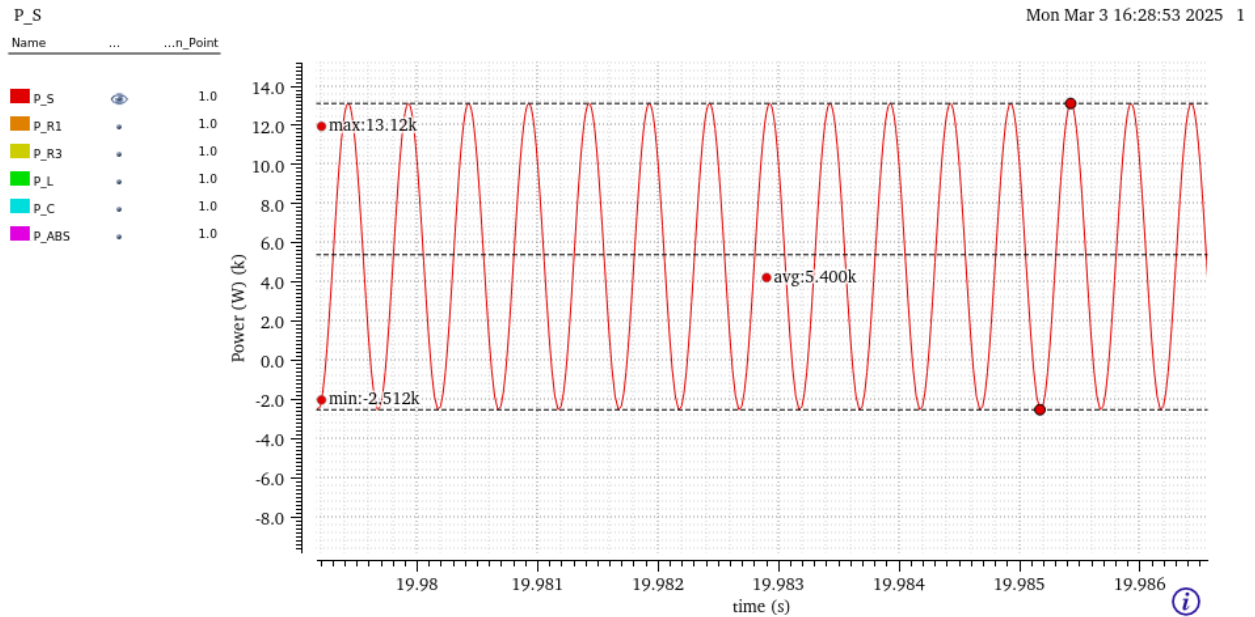


Figure 18: Plot of Power Supplied by the Voltage Source

The magnitude of the average power supplied by the voltage source was found to be 5.400 kW as shown in Figure 18 above.

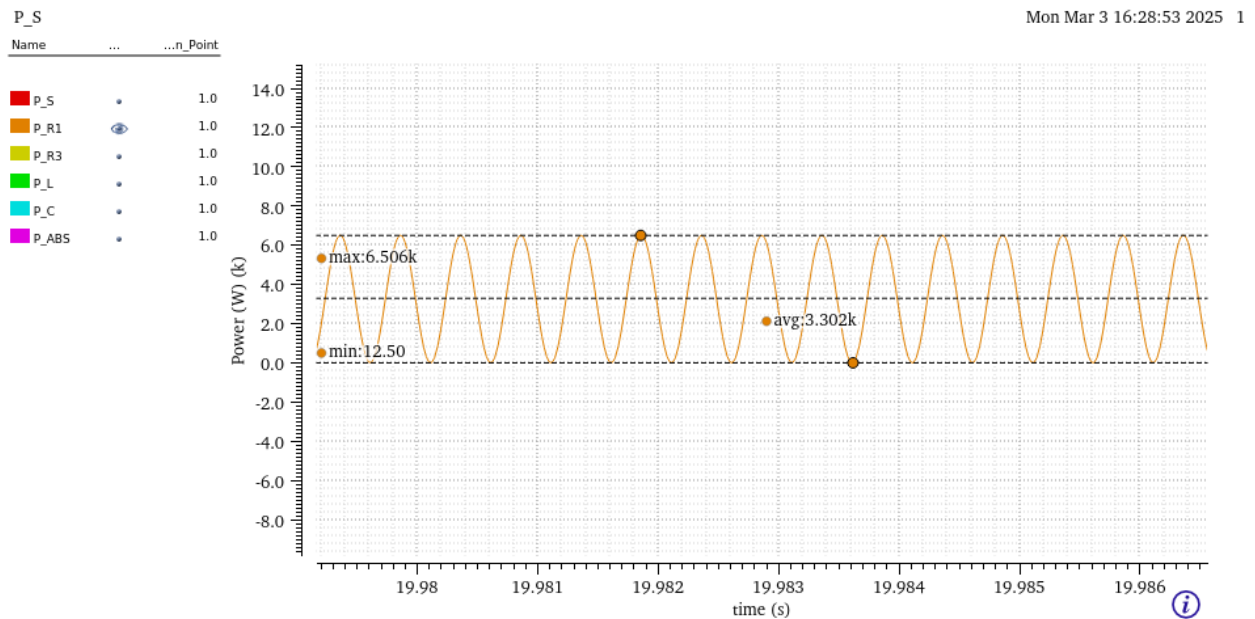


Figure 19: Plot of Power Absorbed by the Resistor (1Ω)

The magnitude of the average power absorbed by the resistor was found to be 3.302 kW as shown in Figure 19 above.

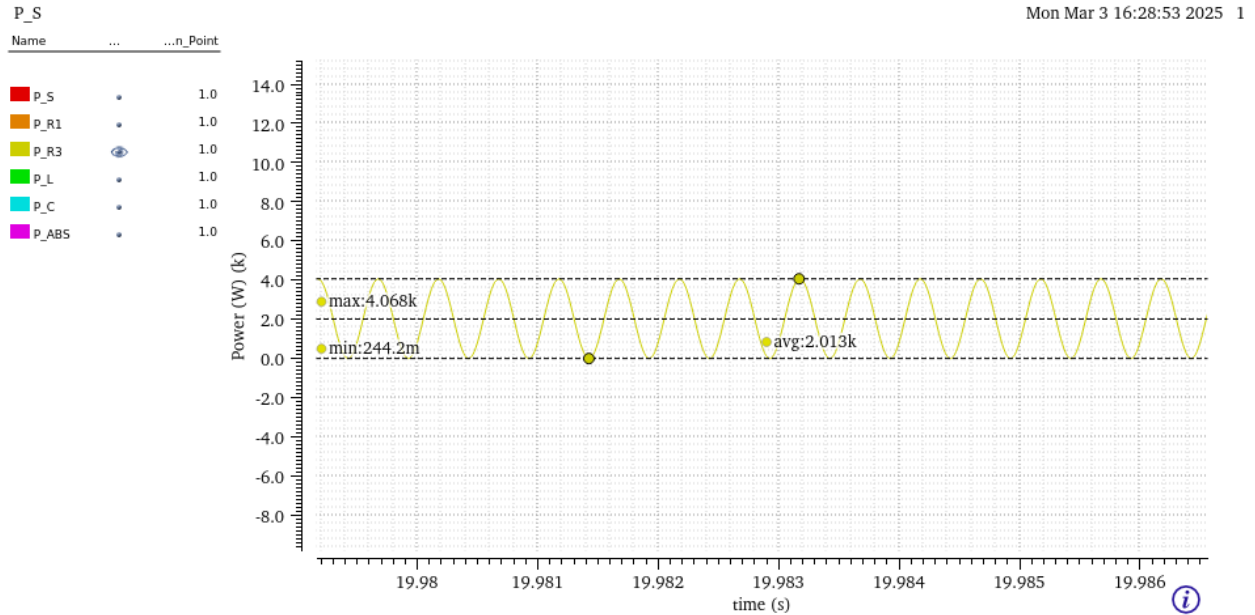


Figure 20: Plot of Power Absorbed by the Resistor (3Ω)

The magnitude of the average power absorbed by the resistor was found to be 2.01 kW as shown in Figure 20 above.

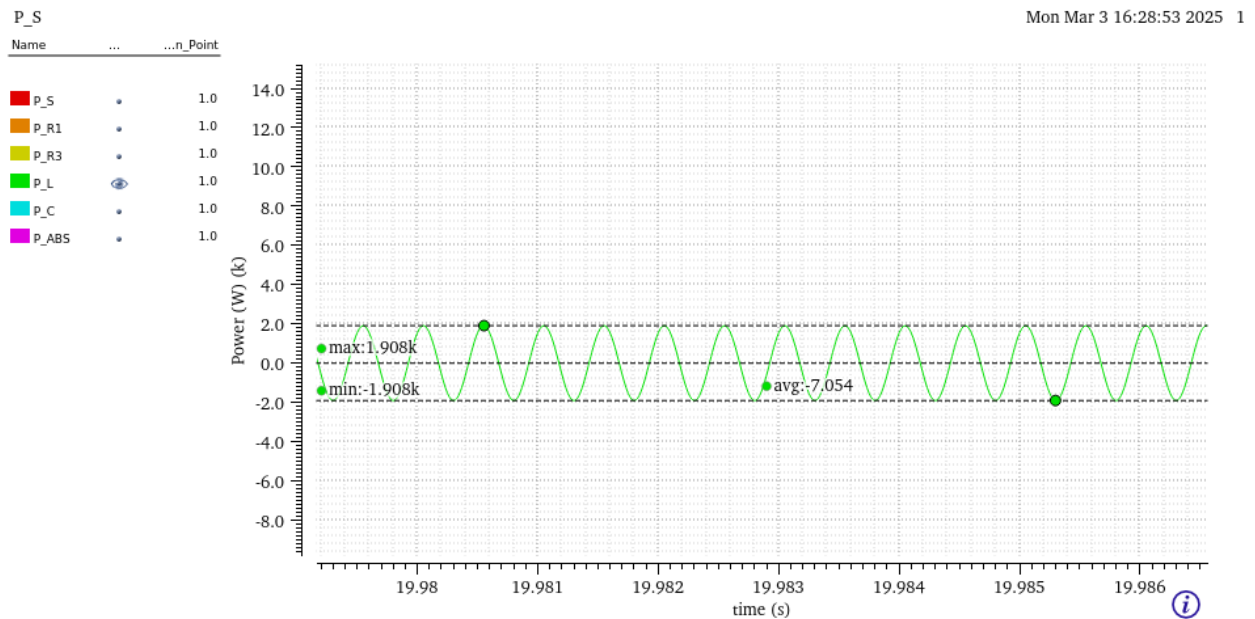


Figure 21: Plot of Power Absorbed by the Inductor

The magnitude of the average power absorbed by the inductor was found to be 7.054 W as shown in Figure 21 above.

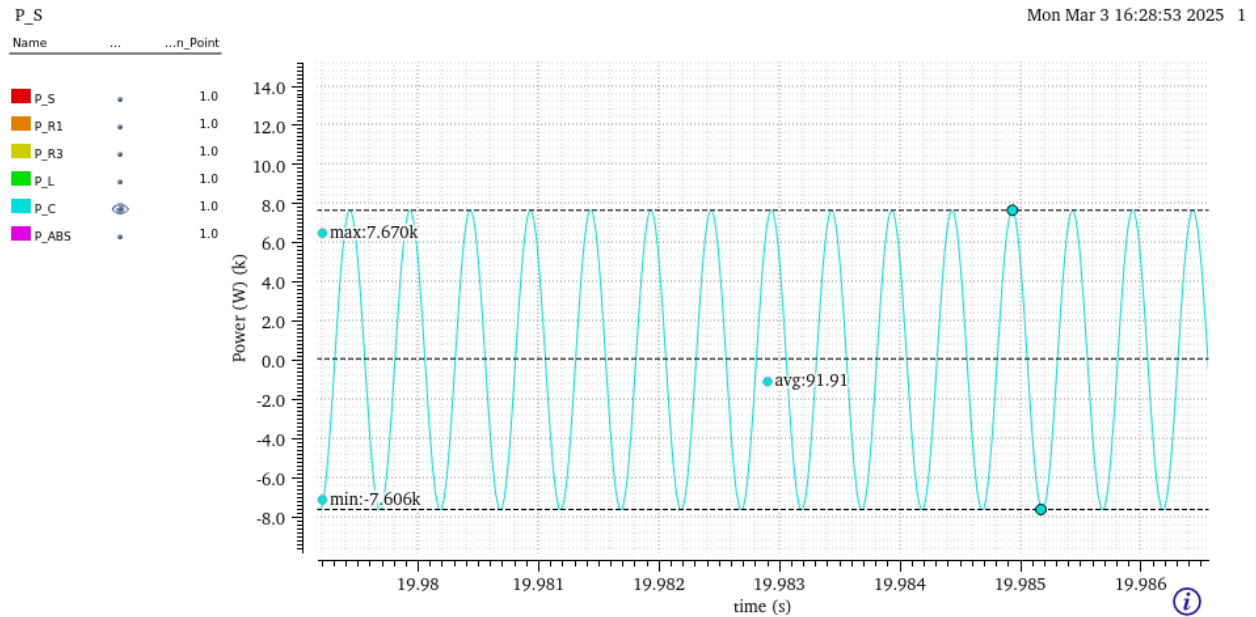


Figure 22: Plot of Power Absorbed by the Capacitor

The magnitude of the average power absorbed by the capacitor was found to be 91.91 W as shown in Figure 22 above.

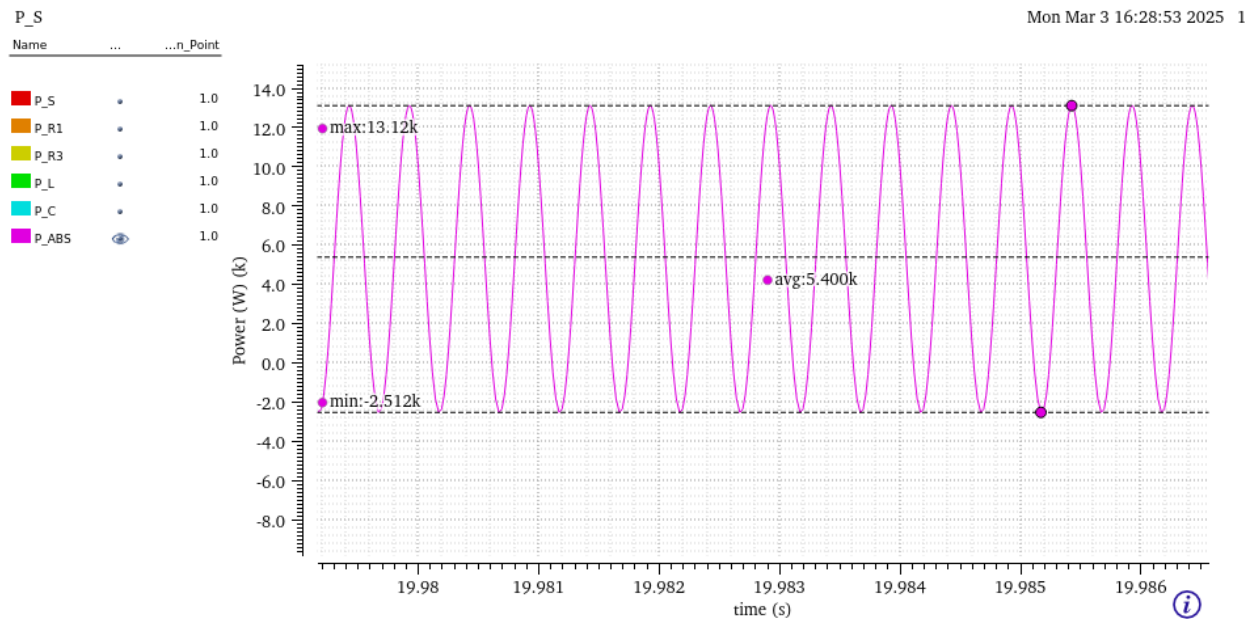


Figure 23: Plot of Power Absorbed by the Resistor

The magnitude of the average power absorbed by the resistor was found to be 5.400 kW as shown in Figure 23 above.

Comparing Figure 18 showing the *Total Average Power Supplied* and Figure 23 showing the *Total Average Power Absorbed*, it can be confirmed that the total average power absorbed and supplied are equal: 5.400 kW.

Discussion of Results:

1. What is your observation about the frequency of the power signal?

From the plots, the frequency of the power signal is observed to be twice the source frequency (2 kHz).

2. Is the power signal's frequency the same as the voltage or current signal frequency?

No, the power signal's frequency is not the same as the source frequency; it has increased by a factor of 2.

3. Is the observed frequency of the power signal expected?

Yes, the observed frequency doubling is expected. Power is given by:

$$P(t) = V(t) \cdot I(t)$$

For an AC circuit:

$$V(t) = V_0 \cos(\omega t), \quad I(t) = I_0 \cos(\omega t + \phi)$$

Multiplying these using the trigonometric identity:

$$\cos A \cos B = 1/2 [\cos(A - B) + \cos(A + B)]$$

yields:

$$P(t) = V_0 I_0 \cos(\phi) + V_0 I_0 \cos(2\omega t + \phi)$$

The first term is a DC component, while the second term is an AC component that described the sinusoidal instantaneous power with twice the original angular frequency (2ω). Since $\omega = 2\pi f$, for 2ω , the frequency becomes $2f$.

4. What is the average power absorbed by the inductor and capacitor? Do you agree with Cadence's results?

In an ideal case, inductors and capacitors should absorb zero average real power since they reactive components and only store/release energy. In the real world, winding resistance in inductors and equivalent series resistance in capacitors may cause small power dissipation. The power of 25.74 W for the inductor and 709.2 mW for the capacitor indicate such losses, while the resistor's 3.166 kW aligns with expected dissipation. Cadence likely accounts for these non-ideal effects, making the results realistic. In a purely ideal circuit, the inductor and capacitor would show 0 W average power instead.

Conclusion

The lab confirmed that the power signal's frequency is twice the source frequency, as expected from theoretical analysis. It was also verified that the average power supplied and absorbed were equal, confirming energy conservation. Additionally, the instantaneous power was plotted, showing its variation over time. While ideal inductors and capacitors absorb zero average power, real-world losses due to resistances in the component cause small power dissipation. The Cadence simulations align with these practical considerations, reinforcing the impact of non-ideal components in circuit analysis.